

```
-- =====
-- MODULE: MANUFACTURING QUALITY & EXECUTION EXTENSION LAYER
-- Database Engine: PostgreSQL \^ Enterprise Configuration
-- File Identity: mfg_quality_and_execution_extension.sql
-- Path: \Database_Design/ERP_Modules_Schema/V_Manufacturing_Production_Module
-- =====
```

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

```
-- =====
-- \. mfg_quality_inspections
-- مدیریت کنترل کیفیت ورودی، تولید و خروجی
-- =====
```

```
CREATE TABLE mfg_quality_inspections (
```

```
    inspection_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    tenant_id UUID NOT NULL,
```

```
    inspection_type SMALLINT NOT NULL,
```

```
-- \: Incoming Material
```

```
-- \: Production Output
```

```
-- \: Final Product
```

```
    source_document_type VARCHAR(100),
```

```
-- INV_DOCUMENT
```

```
-- PRODUCTION_ORDER
```

```
-- PURCHASE_RECEIPT
```

```
    source_document_id UUID,
```

```
    item_id UUID NOT NULL,
```

```
-- Logical Reference -> inv_items
```

```
    batch_id UUID,
```

```
-- Logical Reference -> inv_stock_batches
```

```

inspection_number VARCHAR(100) NOT NULL,

inspection_date TIMESTAMPTZ NOT NULL DEFAULT NOW(),

inspector_user_id UUID NOT NULL,


sample_quantity NUMERIC(10,2) NOT NULL DEFAULT 0,

accepted_quantity NUMERIC(10,2) NOT NULL DEFAULT 0,

rejected_quantity NUMERIC(10,2) NOT NULL DEFAULT 0,


qc_status SMALLINT NOT NULL DEFAULT 1,
-- 1 Pending
-- 2 Approved
-- 3 Rejected
-- 4 Quarantine


notes VARCHAR(500),

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

updated_at TIMESTAMPTZ,

deleted_at TIMESTAMPTZ,

row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE UNIQUE INDEX uq_mfg_quality_inspections_number
ON mfg_quality_inspections(tenant_id, inspection_number)

```

WHERE deleted_at IS NULL;

CREATE INDEX idx_mfg_quality_source
ON mfg_quality_inspections(source_document_type, source_document_id);

-- =====
-- ۲. mfg_work_order_operations
-- اجرای واقعی عملیات تولید
-- =====

CREATE TABLE mfg_work_order_operations (

 operation_execution_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

 tenant_id UUID NOT NULL,

 production_order_id UUID NOT NULL
 REFERENCES mfg_production_orders(production_order_id)
 ON DELETE RESTRICT,

 routing_id UUID NOT NULL
 REFERENCES mfg_production_routing(routing_id)
 ON DELETE RESTRICT,

 work_center_id UUID NOT NULL
 REFERENCES mfg_work_centers(work_center_id)
 ON DELETE RESTRICT,

 operation_sequence INT NOT NULL,

 operator_user_id UUID,

```

planned_start TIMESTAMPTZ,

actual_start TIMESTAMPTZ,

actual_end TIMESTAMPTZ,


planned_hours NUMERIC(ȳ.,ȳ) DEFAULT .,

actual_hours NUMERIC(ȳ.,ȳ) DEFAULT .,


produced_quantity NUMERIC(ȳ.,ȳ) DEFAULT .,

scrap_quantity NUMERIC(ȳ.,ȳ) DEFAULT .,


operation_status SMALLINT NOT NULL DEFAULT \,
-- \ Pending
-- ȳ Running
-- ȳ Completed
-- ȳ Stopped

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

updated_at TIMESTAMPTZ,

row_version BIGINT NOT NULL DEFAULT \
);

```

```

CREATE INDEX idx_mfg_work_operations_order
ON mfg_work_order_operations(production_order_id);

```

```
CREATE INDEX idx_mfg_work_operations_center
ON mfg_work_order_operations(work_center_id);
```

```
-- =====
-- ۳. mfg_material_consumptions
-- مصرف واقعی مواد اولیه در تولید
-- =====
```

```
CREATE TABLE mfg_material_consumptions (

    consumption_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    production_order_id UUID NOT NULL
    REFERENCES mfg_production_orders(production_order_id)
    ON DELETE RESTRICT,

    bom_item_id UUID
    REFERENCES mfg_bom_items(bom_item_id)
    ON DELETE RESTRICT,

    material_item_id UUID NOT NULL,
    -- Logical Reference -> inv_items

    inventory_document_id UUID,
    -- Logical Reference -> inv_documents

    planned_quantity NUMERIC(۲۰,۴) NOT NULL DEFAULT ۰,
```

actual_quantity NUMERIC(٧٠,٤) NOT NULL DEFAULT ٠,

variance_quantity NUMERIC(٧٠,٤)
GENERATED ALWAYS AS
(actual_quantity - planned_quantity)
STORED,

consumption_date TIMESTAMPTZ NOT NULL DEFAULT NOW(),

consumed_by UUID,

status SMALLINT NOT NULL DEFAULT ١,
-- ١ Registered
-- ٢ Posted
-- ٣ Cancelled

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

updated_at TIMESTAMPTZ,

row_version BIGINT NOT NULL DEFAULT ١
);

CREATE INDEX idx_mfg_material_consumption_order
ON mfg_material_consumptions(production_order_id);

CREATE INDEX idx_mfg_material_consumption_item
ON mfg_material_consumptions(material_item_id);